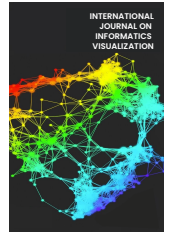




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Engagement Facial Expression Classification Using Landmark-Based Geometric with Bimodal 2-Stream Graph-Based CNN in Online Learning

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Abstract— This study aims to classify facial expressions of engagement using facial landmarks with a combined model between Graph Convolutional Network (GCN) and Convolutional Neural Network (CNN). With this research, researchers hope that this research can be useful for the world of education. In this research, the dataset used comes from the DAiSEE dataset about facial expressions in online learning. The dataset is a video to be extracted into each frame. All of the features in the frame were extracted through 68 landmark points and 52 types of action units. In this research, a comparison was made between GCN architecture and CNN for landmark extraction first to compare the results. This research also uses the LDAM loss function in the combined model to overcome imbalanced datasets. The final results of the model comparison between GCN and CNN using facial landmarks resulted in an accuracy of 50.56% and 49.61%, respectively. The combined GCN and CNN model produces an accuracy of 50.89%. Furthermore, the combined GCN and CNN model using the LDAM loss function produces an accuracy of 52.24%. Although the combined model showed modest performance, it runs significantly lower than temporal models like LSTM and TCN reported in similar studies. Future research could enhance classification accuracy by exploring temporal models more robustly suited to sequence data, such as LSTM or TCN, and by incorporating multimodal inputs to better capture complex engagement expressions in educational contexts.

Keywords— Facial expressions; DAiSEE dataset; engagement; graph convolutional network; convolutional neural network.

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I. INTRODUCTION

The Covid-19 pandemic has significantly changed the learning system. In general, the learning system before the pandemic used a traditional face-to-face system now, it uses an online system [1]. This dramatic shift to online learning has occurred across all levels of education, from primary schools to universities [2]. Although the pandemic is already over, this online learning system has already become widespread, so there are many universities still implementing this learning system [3]. This system utilizes the internet network so teachers and students can connect anywhere and anytime to conduct the teaching and learning process. According to a survey conducted by The Conversation (2020) [4]. Most respondents (50%), including students, lecturers, and university officials, argue that at least 50%- 75% of the total courses can be delivered through e-learning. The survey results show that this e-learning system has a very high potential for implementation in the future.

This rapid shift to online learning has affected teaching practices and fundamentally altered student engagement's nature [5]. By applying this learning system, the interaction between teachers and students becomes more manageable so that students can quickly receive the material the teacher explains. In addition, this learning system provides a new atmosphere in teaching and learning activities, increasing students' motivation to learn the knowledge provided by the teacher [6]. Despite its positive impact, implementing this learning system also has some adverse effects that must be addressed. According to a survey by The Conversation [4], more than 40% of respondents think online learning quality is still poor compared to traditional classroom learning. This is because students and teachers experience several obstacles in the learning process. From the students' perspective, students have difficulty paying attention to the subject matter delivered by the teacher in online face-to-face classes due to unfamiliarity and discomfort in doing online learning, which causes their lack of involvement in the activity. From the

teacher's perspective, teachers find it challenging to recognize their students' level of interest or learning engagement [7], [8]. Recent studies have begun to analyze the effectiveness of online learning during the pandemic on a global scale. Bond et al. [9] conducted a comprehensive study examining the first international online semester across multiple countries, while García-Morales et al. [10] investigated the transformation of higher education in the online learning scenario, and Cao et al. [11] conducted research about the psychological impact on students during the online learning transition. These findings are further supported by Marinoni et al. [12], who conducted a global survey revealing widespread challenges in maintaining student engagement across different geographical and cultural contexts, collectively emphasizing the urgent need for better engagement detection tools. To overcome this, it is necessary to create a system that can recognize the level of student involvement in the online learning process through facial expressions.

Facial expressions are a type of non-verbal communication in humans that can convey a person's emotional state to others who are observing it. Through facial expressions, we can understand the emotions that are being felt in a person [13], [14]. These expressions can arise when a person experiences certain emotions that result in facial muscle movements. This difference in emotion affects the difference in muscle movement on the face, which can eventually lead to various kinds of expressions [15]. These facial expressions can be sad, happy, angry, and other. These emotions have a significant role in our cognitive and behavioral conditions. In the world of education, students' emotional state can affect them in terms of their concentration and motivation [16]. In the online face-to-face learning process where a teacher is not present in person, facial expression recognition is essential for teachers to see the emotional state of their students, such as whether the student can follow this lesson well. The process of recognizing emotions on this face was identified based on changes in critical facial features. These features include eyes, eyebrows, mouth, and forehead [17]. The following are some previous studies on facial expression recognition.

The first research paper by [18] has used the DAiSEE dataset, InceptionNet, 3D CNN, and LRCN for the model's architecture. The research was conducted in all classes on the DAiSEE dataset: boredom, engagement, confusion, and frustration. This study's results resulted in the highest accuracy value in the LRCN model, with the results being 53.7%, 57.9%, 72.3%, and 73.5%, respectively.

The second research paper by [19] compared two facial feature extraction methods, namely facial landmarks and facial embedding. Both of these extraction methods were trained using LSTM. The dataset used in this study is the DAiSEE dataset in the boredom and frustration classes. The results of this study on the F1-score metric with the facial landmark method in the boredom and frustration classes are 0.696 and 0.419. Meanwhile, the F1-score metric for the facial embedding method in the boredom and frustration classes is 0.56 and 0.356.

The third research paper in [20], this research uses the DAiSEE dataset on the engagement class and the model used is the Temporal Convolution Network combined with bidirectional LSTM. This research also used a combination of several feature extractions, namely eye gaze direction, eye

coordinates, face coordinates, rigid face, non-rigid face, and facial action units. The results of this study using Temporal Convolution combined with bidirectional LSTM resulted in an accuracy of 60%, and the combined use of all these feature extractions resulted in an accuracy of 59%.

The fourth research paper in [21], this research used the DAiSEE dataset on the engagement class and performed image classification with a single frame using 10 intervals on the dataset. The model is a vision transformer with several loss functions to overcome imbalanced datasets. The loss functions used include Categorical Cross Entropy loss, Class Balanced Loss, and Focal Loss. The result of this research with Categorical Cross Entropy loss function is 52.16%, Class Balanced Loss function with Categorical Cross Entropy loss is 55.18%, Focal Loss function is 54.78%, Class Balanced Loss function with Focal Loss is 52.44%, Weighted Categorical Cross Entropy loss function is 49.86%, and Weighted Focal loss function is 51.83%.

In the fifth research paper by [22] has used DAiSEE and VRSEE datasets from the engagement class. It classified images by sequencing using 60 frames in each video. Three models are used: EfficientNet B7 + TCN, EfficientNet B7 + LSTM, and EfficientNet B7 + Bidirectional LSTM models. This study's accuracy results on each of the EfficientNet B7 + TCN, EfficientNet B7 + LSTM, and EfficientNet B7 + Bidirectional LSTM models on the DAiSEE dataset are 64.67%, 66.39%, and 67.48%, respectively.

The sixth research paper by [23] has used a combined dataset between JAFFE, FER2013, and CK+. This research used landmark extraction converted into a Delaunay triangulation graph and trained using the Graph Convolutional Network model. The final result of this research produces an accuracy rate of 95.85% on the combined three datasets.

The research carried out used all data on the engagement class from the DAiSEE dataset created by [18]. For feature extraction, the methods used were facial landmarks [19], and action units [20]. Then, in research [21], several loss functions were used to overcome imbalanced datasets, but none could overcome the imbalanced dataset. Therefore, researchers want to use the LDAM loss function proposed by [24] to overcome imbalanced datasets. In research by [23], facial expression recognition was conducted using GCN architecture along with the Delaunay triangulation graph, and pretty good results were obtained. Therefore, researchers used GCN architecture and Delaunay triangulation graphs to process landmark data. In research by [22], EfficientNet architecture with the LSTM and TCN models were combined to improve accuracy results. Therefore, the researchers want to use the GCN model merger with CNN to improve accuracy results. In addition, in the research paper by [22], the training process used sequences through the LSTM model to produce good accuracy. Therefore, researchers used sequences by aggregating the SoftMax values of all classes in all frames in the same video.

Based on this explanation, this research aims to recognize students' level of engagement or interest during teaching and learning activities. This research used the DAiSEE dataset, which contains videos of human facial expressions, as the primary material. This research is expected to help teachers determine the level of involvement or interest of the students

they are teaching so that they can provide the feedback expected by the teacher.

II. MATERIALS AND METHOD

In this research, six stages were carried out, as shown in Fig 1 below.

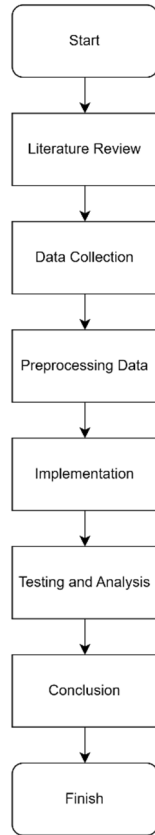


Fig. 1 Research Stages

A. Literature Review

At this stage, the aim is to find and review previous studies to be used as references for this research.

B. Data Collection

This stage aims to find the data needed to conduct this research. In this study, the data used is secondary data from the DAiSEE dataset. This dataset contains videos of human facial expressions in a virtual environment consisting of 9068 videos divided into four classes: Boredom, Engagement, Confusion, and Frustration. Each class has four subclasses: Very Low, Low, High, and Very High. These large-scale datasets play a crucial role in improving the accuracy and robustness of emotion detection systems. Li and Deng [25] emphasize that these comprehensive datasets, coupled with advanced deep learning architectures, are fundamental to achieving state-of-the-art performance in facial expression recognition. This research only uses the engagement class in all data and uses the same train, validation, and test split as the DAiSEE dataset.

C. Preprocessing Data

This stage aims to preprocess the data before it is used. In this research, the DAiSEE dataset was extracted into each frame. One video in this dataset produced approximately 300

frames. After that, all frames would be extracted features using facial landmarks with 68 points and action units. The facial landmark data consists of 2 data preprocessing stages that would be carried out sequentially. In the first stage, the data would be normalized using MinMax Normalization. In the last stage, the normalized data would be converted into a graph using the Delaunay Triangulation technique. In the action unit data, the data is only normalized using MinMax Normalization.

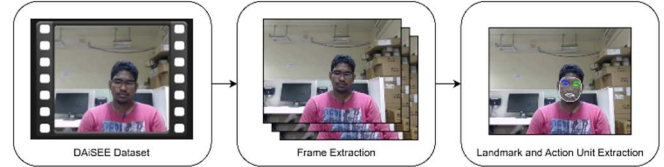


Fig. 2 Preprocessing Stage

D. Implementation

This stage aims to process data that is ready to be used for training using a predetermined model. This research used three model architectures: the Graph Convolutional Network (GCN) architecture model, the Convolutional Neural Network (CNN) architecture model, and the combined GCN and CNN architecture model. The GCN and CNN architecture models were used as input data in the form of landmarks. Meanwhile, the combined GCN and CNN architecture uses landmark data and action units.

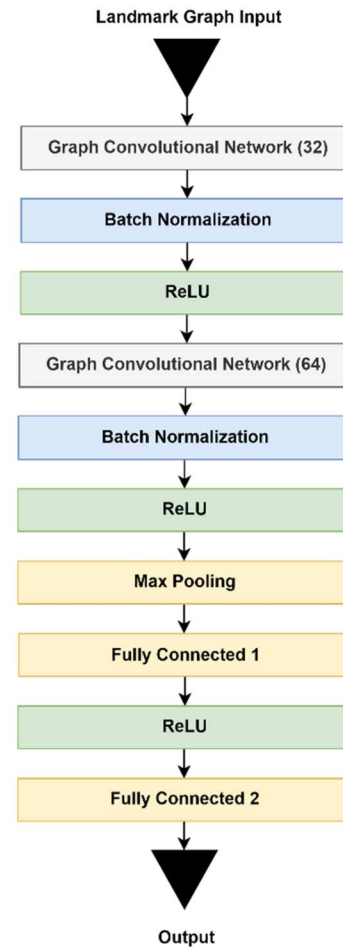


Fig. 3 GCN Model Architecture

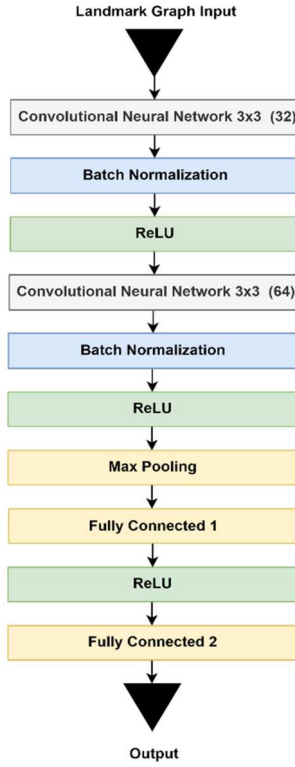


Fig. 4 CNN Model Architecture

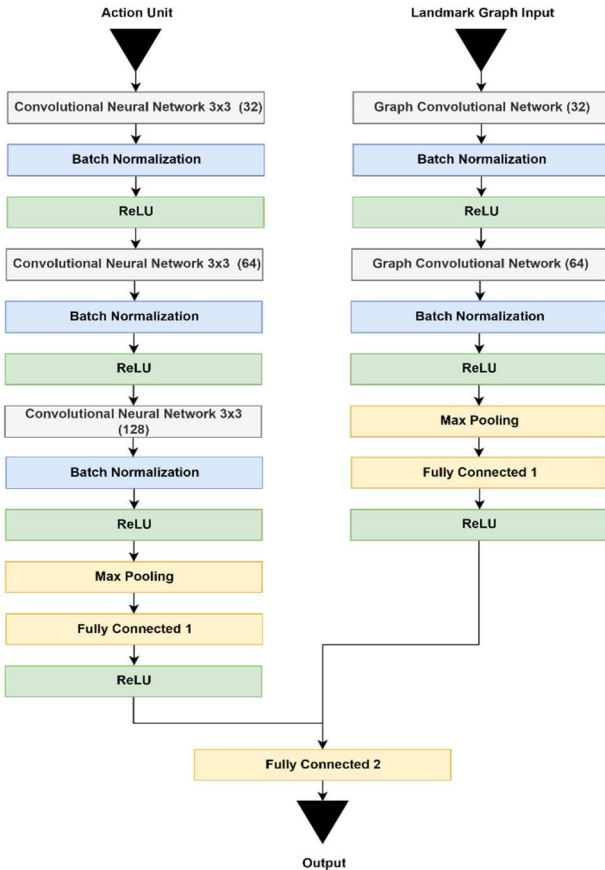


Fig. 5 Combined Model Architecture

E. Testing and Analysis

This stage aims to compare the architecture of GCN and CNN models in processing landmark data. The two models

are compared to find the best architecture in training landmark-shaped data. After that, a combination of GCN and CNN architecture was used to improve accuracy. In this model, there are two different input data for each architecture. GCN architecture uses landmark data. Meanwhile, the CNN architecture uses Action Unit data. In the last test, we would use the LDAM loss function to overcome the imbalanced dataset in the combined model.

1) *Convolution Neural Network (CNN)*: CNN is a neural network architecture that effectively manages data with coordinate shapes such as landmarks. This is because CNN can correctly anticipate landmarks by learning the structural characteristics of the image [26]. This architecture has shown significant success in computer vision tasks, particularly in facial analysis applications [27].

2) *Graph Convolutional Layer (GCN)*: GCN is an efficient variant of Convolutional Neural Network (CNN) that can only operate on graphs. This architecture allows each node in its graph to borrow features from its neighbors and use them to combine its features. This is done to enrich the nodes' embeddings so they can be studied more fully [28]. In the recent research by Liao [29], the GCN architectures have shown particular promise in capturing both spatial and temporal aspects of facial expressions.

3) *Label Distribution Aware Margin (LDAM) Loss*: LDAM Loss is a loss function designed to address imbalanced datasets. This loss function works by increasing the margin for the minority class to improve the resulting performance for that class [24]. The equation of this function can be seen as follows:

$$\mathcal{L}_{\text{LDAM}}((x, y); f) = -\log \frac{e^{z_y - \Delta y_i}}{e^{z_y - \Delta y_i} + \sum_{j \neq y} e^{z_j}} \quad (1)$$

$$\Delta y_i = \frac{C}{n_{y_i}^{1/4}} \quad (2)$$

Where:

- x_i = input data
- y_i = label
- z_j = logit value for class j
- Δ_{n_i} = margin value for each class
- C = scalling factor
- n_{y_i} = total data for class y_i

III. RESULTS AND DISCUSSION

At this stage, we present the research results and its explanation. The data studied in this study comes from the DAiSEE dataset in the engagement class, which consists of 4 class levels, namely 0, 1, 2, and 3. The distribution of data in each class can be seen in Table 1 as follows:

TABLE I
DATA DISTRIBUTION FOR EACH CLASS

Class	Total Data
0	61 videos
1	459 videos
2	4477 videos
3	4071 videos
Sum Total	9068

A. Comparison of GCN and CNN Architecture Accuracy on Landmark Extraction Method

Testing was carried out at this stage to find the best architecture for managing landmark data. Testing was only be carried out on GCN and CNN architectures using the same parameters: learning rate = 0.001, batch size = 32, and Cross-Entropy loss function. The accuracy result obtained in the GCN model is 50.56%, while the CNN model gets an accuracy result of 49.61%. The results of this test are shown in Table 2.

TABLE II
GCN AND CNN MODEL COMPARISON RESULTS

Model	Accuracy Result
GCN	50.56%
CNN	49.61%

Based on the data in Table 2, the model's performance with GCN architecture can work better than the CNN architecture model. The difference in accuracy results occurs because the model with GCN architecture can learn features on landmark graph data using Delaunay Triangulation better than the CNN architecture model with landmark data. The confusion matrix results on the GCN and CNN models are displayed in Fig 6 and Fig 7 as follows.

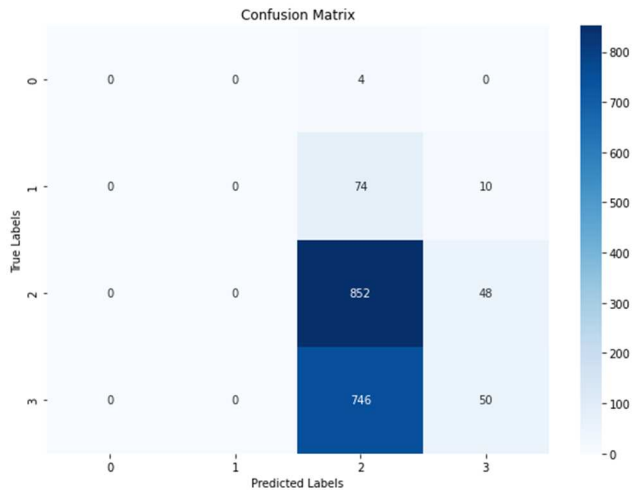


Fig. 6 GCN Model Confusion Matrix

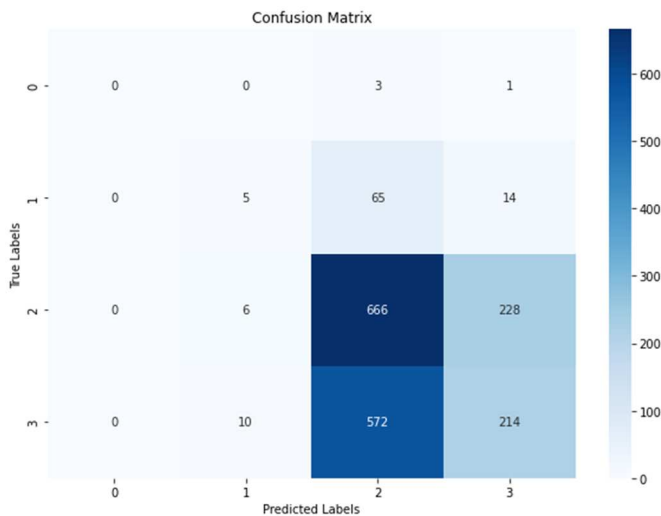


Fig. 7 CNN Model Confusion Matrix

These two models can only recognize data from classes 2 and 3. They still fail to recognize data from the minority classes, namely classes 0 and 1. One reason this happens is the imbalanced dataset in classes 0 and 1.

B. Combining GCN Architecture with CNN Architecture (2-streams)

At this stage, we combined the architecture between GCN and CNN using two different input types. In the GCN model, the data used is landmark data, which is converted into a Delaunay Triangulation Graph. In the CNN model, the data used is Action Unit data. This test's parameters are learning rate = 0.001, batch size = 32, and Cross-Entropy loss function. The combined model's accuracy result is 50.89%. The results of this test are shown in Table 3.

TABLE III
COMBINED MODEL RESULTS

Model	Accuracy Result
GCN	50.56%
CNN	49.61%
GCN+CNN	50.89%

Based on the data in Table 3, the model's performance with the combined GCN and CNN architecture can be slightly better than that of the GCN model. This increase in accuracy may be due to the use of action unit data in the CNN model. The confusion matrix results for combining the GCN model with CNN are displayed in Fig 8.

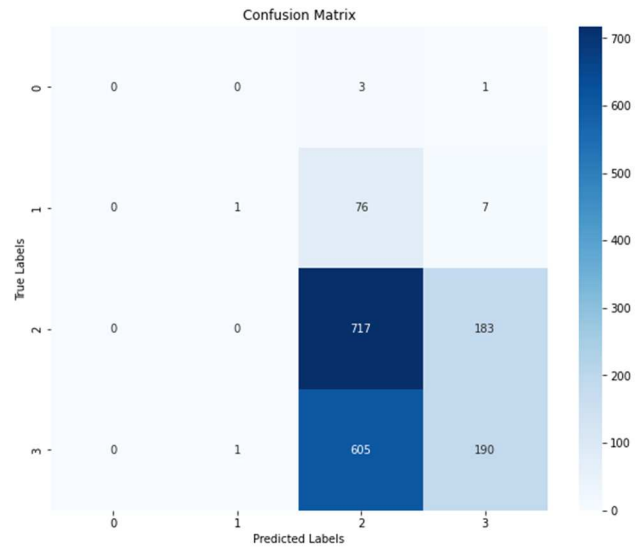


Fig. 8 Combined Model Confusion Matrix

This model is similar to the previous two, which can only recognize class 2 and class 3 data. In the minority classes, namely classes 0 and 1, these models still fail to recognize data in these classes. One reason this happens is the imbalanced dataset in classes 0 and 1.

C. Use of LDAM Loss Function in Combined Model

Testing was carried out at this stage using the LDAM loss function on the combined model. This is done to overcome the imbalanced DAiSEE dataset. In this test, the parameters used are learning rate = 0.001, batch size = 32, and LDAM loss function. The prediction results obtained correctly in

classes 0, 1, 2, and 3 using the LDAM loss function are 0, 1, 657, and 273, respectively. While in the model that only uses the usual cross-entropy loss function in classes 0, 1, 2, and 3 are 0, 0, 764, and 140, respectively. The results of this test are displayed in Table 4.

TABLE IV
COMPARISON OF LOSS FUNCTIONS IN COMBINED MODELS

Loss Function	Label 0	Label 1	Label 2	Label 3	Accuracy
CrossEntropy	0	0	764	140	50.89%
LDAM	0	1	657	273	52.24%

Based on the test results of using the LDAM loss function on the combined GCN and CNN architecture on all classes and accuracy results in Table 4, the accuracy value is 52.24%. Based on this data, the model's performance with the combination of GCN and CNN architecture with the LDAM loss function can work slightly better than the combined GCN and CNN model with the cross-entropy loss function. The increase in accuracy results is due to using the LDAM loss function. Based on the performance results of the model, even though it has used the LDAM loss function which aims to overcome the imbalanced dataset, the model still has difficulty in recognizing patterns in minority classes. The results of the confusion matrix combining the GCN model with CNN using the LDAM loss function are shown in Fig 9 as follows.

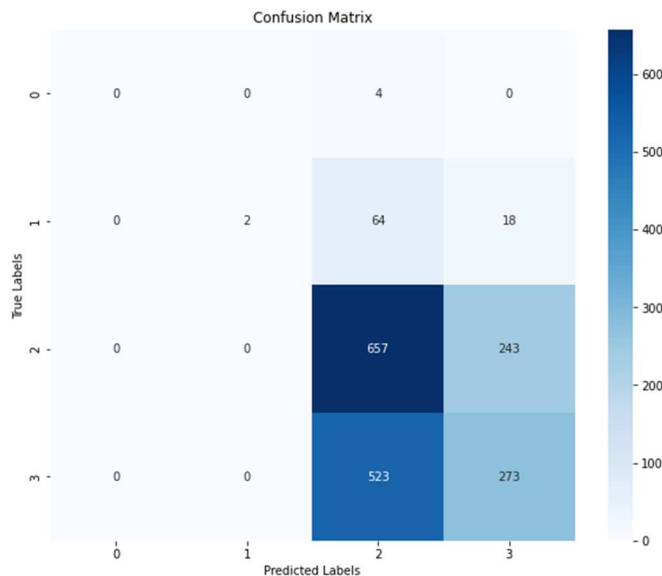


Fig. 9 Combined Model with LDAM Loss Confusion Matrix

This model is similar to the previous three models, which can only recognize class 2 and class 3 data. In the minority classes, namely classes 0 and 1, these models still do not recognize data in these classes despite using the LDAM loss function.

D. Comparison of Previous Research

Based on the results of the research, four types of models were produced. These models are models using GCN, models using CNN, models using a combination of GCN and CNN, models using a combination of GCN and CNN, and models using a combination of GCN and CNN plus the LDAM loss

function. Of the four models, the model with a combination of GCN and CNN plus the LDAM loss function produces the following accuracy: 49.61%, 50.56%, 50.89%, and 52.24%. Then, a comparison was made with the previous research to find out the increase in accuracy in previous studies. This comparison are listed in Table 5.

TABLE V
COMPARISON RESULTS WITH PREVIOUS RESEARCH

References	Method	F1 score	Accuracy
[18]	InceptionNet (Frame Level)	-	46.40%
[18]	InceptionNet (Video Level)	-	47.10%
Ours	CNN	0.46	49.61%
Ours	GCN	0.38	50.56%
Ours	GCN+CNN	0.46	50.89%
Ours	GCN+CNN+LDAM Loss	0.49	52.24%
[30]	Inflated 3D ConvNet	-	52.35%
[31]	ResNet+TCN (weighted loss)	-	53.70%
[21]	ViT	0.49	55.18%

IV. CONCLUSION

Based on the research results, several conclusions can be drawn. Using the GCN model architecture for feature extraction using landmarks results in better model performance than the CNN model architecture. Specifically, the GCN model achieved an accuracy of 50.56%, while the CNN model only achieved an accuracy of 49.61%. Additionally, combining the GCN model architecture with CNN improves model performance in recognizing engagement facial expressions on the DAiSEE dataset. This combined model achieved an accuracy of 50.89%.

Furthermore, the LDAM loss function did not significantly enhance the model's performance in addressing minority classes. However, it did lead to an overall increase in accuracy. The model utilizing the LDAM loss function achieved an accuracy of 52.24%, representing an improvement of 1.57% over the combined GCN and CNN model architecture without the LDAM loss function. These findings suggest that while GCNs outperform CNNs in this context, combining both architectures yields slight improvements. The LDAM loss function offers a more substantial accuracy boost even though it may not effectively address class imbalances.

Engagement detection is key to improving online learning experiences. Xie et al. [32] revealed that understanding and improving student engagement is essential in post-pandemic education's 'new normal' and 'next normal'. So, several suggestions can be made to improve the model's performance in this study. Firstly, during the extraction stage from video to frame on the DAiSEE dataset. It is advisable to use intervals instead of all existing frames. This approach could enhance the model's accuracy and reduce the computation time. Exploring model architectures that can leverage spatiotemporal data, such as TCN (Temporal Convolutional Networks) and LSTM (Long Short-Term Memory Networks), might provide additional performance benefits. Lastly, performing hyperparameter tuning is recommended to further optimize the model's performance. By adjusting and fine-

tuning the hyperparameters, it is possible to achieve more accurate and reliable results.

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